

ASSIGNMENT: 09

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Roll No: CSB25203

Subject: Advanced Programming

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import java.util.ArrayList;

// Base Class
class Account {
    private String accountNumber;
    private String ownerName;
    private double balance;

    // Constructor chaining
    public Account() {
        this("000", "Unknown", 0.0);
    }

    public Account(String accountNumber, String ownerName, double balance) {
        this.accountNumber = accountNumber;
        this.ownerName = ownerName;
        setBalance(balance);
    }

    // Getters & Setters (Encapsulation)
    public String getAccountNumber() {
        return accountNumber;
    }

    public String getOwnerName() {
        return ownerName;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        if (balance < 0) {
            throw new IllegalArgumentException("Balance cannot be negative");
        }
        this.balance = balance;
    }
}
```

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    }

    // Deposit
    public void deposit(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid deposit amount");
        }
        balance += amount;
    }

    // Withdraw
    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid withdrawal amount");
        }
        if (amount > balance) {
            throw new IllegalArgumentException("Insufficient balance");
        }
        balance -= amount;
    }

    // Display
    public void display() {
        System.out.println("Account No: " + accountNumber);
        System.out.println("Owner: " + ownerName);
        System.out.println("Balance: " + balance);
    }
}

// Savings Account
class SavingsAccount extends Account {
    private double interestRate;

    public SavingsAccount(String accNo, String name, double bal, double rate)
    {
        super(accNo, name, bal);
        this.interestRate = rate;
    }

    @Override
    public void display() {
        super.display();
        double interest = getBalance() * interestRate / 100;
        System.out.println("Interest Rate: " + interestRate + "%");
        System.out.println("Interest: " + interest);
    }
}

```

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// Current Account
class CurrentAccount extends Account {
    private double overdraftLimit;

    public CurrentAccount(String accNo, String name, double bal, double limit)
    {
        super(accNo, name, bal);
        this.overdraftLimit = limit;
    }

    @Override
    public void withdraw(double amount) {
        if (amount <= 0) {
            throw new IllegalArgumentException("Invalid withdrawal");
        }
        if (amount > getBalance() + overdraftLimit) {
            throw new IllegalArgumentException("Overdraft limit exceeded");
        }
        setBalance(getBalance() - amount);
    }

    @Override
    public void display() {
        super.display();
        System.out.println("Overdraft Limit: " + overdraftLimit);
    }
}

// Main Class
public class BankingSystem {
    public static void main(String[] args) {

        ArrayList<Account> list = new ArrayList<>();

        // Polymorphism
        list.add(new SavingsAccount("101", "Rajdeep", 10000, 5));
        list.add(new CurrentAccount("102", "Aman", 5000, 2000));

        for (Account acc : list) {
            try {
                acc.deposit(1000);
                acc.withdraw(3000);
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            }

            System.out.println("\n--- Account Details ---");
            acc.display();
        }
    }
}
```

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}  
  }  
}
```